

INTRODUCTION TO INTERNET (COM 122)

LECTURES NOTE

**COURSE LECTURER:
SHEHAHYEL, JOSHUA**

**DEPARTMENT OF COMPUTER SCIENCE
FEDERAL POLYTECHNIC MUBI**

CONCEPT OF INTERNET

The **Internet** is the global system of interconnected computer networks that use the Internet protocol suite (TCP/IP) to link devices worldwide. It is a *network of networks* that consists of private, public, academic, business, and government networks of local to global scope, linked by a broad array of electronic, wireless, and optical networking technologies. The Internet carries a vast range of information resources and services, such as the inter-linked hypertext documents and applications of the World Wide Web (WWW), electronic mail, telephony, and file sharing.

INTERNET

The Internet commonly referred to as the „Net“ is better described than defined, as a global network of computers. It is often described as a network of networks because it first began through the linking of the existing local computer networks used by universities and governmental organizations. The Internet is a constellation of computers around the world that speak the same language so information travels seamlessly from one computer to another. Globally, the Internet connects scientific research, educational, government, commercial and business networks. When you connect to the Internet, your computer becomes part of the global network of computers. The Internet is more than a technological marvel. The pace is quick, with messages and information racing round the world in a few seconds. It has ushered in an era of sweeping changes in networking and communication that has left no business, agency, organization or government untouched.

How the Internet Developed

The Internet began in the 1960s when the US Department of Defense through ARPANET (Advanced Research Project Agency Network) began creating a computer network that would continue to function in the event of a major disaster. In 1969, four mainframe computers at a U.S. university were linked into a network that transferred data on dedicated high-speed transmission lines. Through the 1970s and 1980s new similar networks were created to link educational institutions, government offices and military centers in the United States and other countries. In the 1990s, as the Internet grew, private businesses and other regional network providers took over the operation of the major Internet networks and the development of much of the information available online. Private businesses, educational institutions, organizations and individuals

now develop much of the information content available online.

How Information is transmitted on the Internet

The Internet is a worldwide complex of small regional networks. To understand this, picture a major road connecting large cities. Smaller roads link the cities to small towns where residents travel on narrow residential streets. The high-speed Internet is the major road in this scenario. Connected to it are smaller computer networks (functioning like smaller, less traveled roads) that can share data information at high speeds. In order to enable different computers to communicate and transmit packaged information with each other, a standard „protocol“ has been established for transporting data. This Internet protocol is called Transmission Control Protocol/Internet Protocol (TCP/IP) and allows all networks all over the world to communicate with each other on the Internet. When transmitting information over a network or the Internet, it is in the same binary form as when it is stored on a computer, but the data has to be packaged for transmission. This procedure is necessary because bits travelling freely on the network will have no meaning to network devices. So before data is transmitted on the Internet, network software forms the data into packets.

Benefits of the Internet

Basically and generally, the benefits derivable from the utilization of the Internet are mainly as follows:

- Speed/Time Saving:** The speed of transmission on the Internet is the significant benefit of the Internet. A letter or document that could take days to arrive by regular mail can be sent to the other side of the world in minutes. Likewise, searching through a card catalogue at a library can be a time consuming, tedious process, but you can search an online version of the same catalogue in a fraction of the time. The speed of transmitting information naturally saves time and money.
- Breadth/Reach:** The Internet gives you access to a vast (and growing) collection of databases, documents, computer software; these and almost any other kind of information can be stored electronically.
- Cost Saving:** Exchanging information via the Internet is less expensive than using telephones or fax machines especially where telephone access fees are high. Likewise marketing and advertising your products and services on the Internet can be less expensive than using conventional printing and document-delivery methods.
- Two-way Communication:** Audiences now have the means to respond directly

to sources of information, research and opinion.

•**Trading:** The Internet provides the opportunity for purchase and sale of goods and services. Specifically, the Internet through the E-Commerce has provided the business community, among others, the following benefits:

- i. Improved response time to clients' requests
- ii. Improved competitive position
- iii. Eased the process of concluding deals and financial transactions
- iv. Extended market reach and increased revenue potentials
- v. Increased consumer convenience and choice
- vi. Reduced prices
- vii. Improved customer service

SERVICES OFFERED ON THE INTERNET

The major Internet services are as follows:

Electronic Mail (e-mail): This is the most popular and most utilized Internet service especially in the developing countries of the world. A conservative estimate puts the number of people using the e-mail worldwide at more than 90 million people. E-mail is message sent from one person to another via a computer network. The same e-mail can be sent easily to one address or many addresses. Commercial electronic mail is used for the same purposes as the organization wide mail system. But instead of being limited to members of one organization, the service is made available to the public on subscription bases.

E-mail messages are exchanged between customers, suppliers, friends and business partners. E-mail service has cut down on the cost of mailing or sending information thus saving time and money. E-mail is used as a marketing tool, for example, a company, Dealaday.com uses e-mail to prospect for customers and sends group e-mail to 10,000 user names all customers who have indicated they would like to receive e-mails. With e-mail they receive a 7%, 8% and sometimes 10% response, unlike 1% or 2% response in normal direct marketing.

The World Wide Web (WWW): The World Wide Web or the web is one of the Internet's most popular applications and it was launched in 1991. It is a graphical, easy-to-use way to organize and present information, including texts, images, movies, sounds and more.

The World Wide Web has generated tremendous popular interest in the Internet. The www is unique for two reasons. Firstly it is highly interactive media bringing documents in graphics, audio and video. Secondly, it uses hyperlink, which

provides connections between different resources. It allows users to jump from one page to another.

A file in www is called the home page and usually contains a multimedia clip. A page can link you to other web pages and Internet resources with clicks of a mouse. You can view and download any information you need on any home page with the aid of software known as the web browser.

File Transfer Protocol (ftp): File transfer protocol (ftp) is an Internet tool used to transfer files between computers and it is the most common method of transferring files on the Internet. Without viewing them as they are transferred the ftp enables you to access file on a remote location on the Internet once you log on to an ftp site, and you are able to access all files on sites and download them if you so desire. It is fairly easy to publish information on the Internet. Many institutions maintain publicly accessible archive of information that they want to share with others.

Telnet: Telnet is an Internet service that allows you to connect to a remote computer to use specific databases or other applications available on that computer. Telnet is one of the first applications widely used on the Internet. Many telnet applications are now available on the World Wide Web where they are easier to use.

Internet Conference: This is an Internet service whereby different techniques are used to allow people to discuss topics of mutual interest. A conference can be as simple as sending e-mail to many different people or as complex as arranging to have people link different cities, see images of each other and hear each other talk in real time- that is video conferencing. Internet conferencing is possible, but the technology becomes more complex and the speed of the connection becomes more important as you go from text only, to transmitting sound and video in real time. Internet conference brings about tremendous savings on travelling, time and money.

Internet Telephony: This is a service in which you can use the Internet as a voice telephone line. With some types, both you and the person you are calling must have an Internet connection and be online at the time you wish to talk. Some companies also offer Internet-to- telephone services where from your computer you can call any telephone number in the world. Internet telephone services are inexpensive when you compare them with the conventional cost of telephone services. If your organization spends a lot of money on long distance telephone calls, this might be a good option for you. There are growing numbers of

companies providing Internet telephone services.

Internet Fax: Internet fax is a service that takes a special type of e-mail message and sends it to a fax machine specified in the message. The message can be faxed to a computer (Internet-to-Internet).

Intranet and Extranet

Intranet

An intranet is a local area network that companies use to distribute information and speed up the movement of data within offices. In other words an intranet allows the job of processing to be distributed among multiple computers. Intranet activities usually take place behind secure „firewalls“ that only authorized users have access to.

An intranet can span multiple business locations via the Internet. Intranets are built using the same standard Internet software such as TCP/IP, e-mail client, web browser and telephone dialer.

Advantages of Intranets

Workforce productivity: Intranets can help users to locate and view information faster and use applications relevant to their roles and responsibilities.

Time: With intranets, organizations can make more information available to employees on a "pull" basis (i.e. employees can link to relevant information at a time which suits them) rather than being deluged indiscriminately by emails.

Communication: Intranets can serve as powerful tools for communication within an organization, vertically and horizontally.

Web publishing allows 'cumbersome' corporate knowledge to be maintained and easily accessed throughout the company using hypermedia and Web technologies. Examples include: employee manuals, benefits documents, company policies, business standards, news feeds, and even training, can be accessed using common Internet standards (Acrobat files, Flash files, CGI applications). Because each business unit can update the online copy of a document, the most recent version is always available to employees using the intranet.

Cost-effective: Users can view information and data via web-browser rather than maintaining physical documents such as procedure manuals, internal phone list and requisition forms.

Extranet

When a company makes open its internal network or intranet to selected business partners, the intranet becomes an extranet. Suppliers, distributors and

other authorized users can then connect to the company's network over the „Net“ or through virtual private networks. Once inside, they can view the data the company makes available.

Advantages of Extranet

- Exchange large volumes of data using Electronic Data Interchange (EDI)
- Share product catalogs exclusively with wholesalers or those "in the trade"
- Collaborate with other companies on joint development efforts
- Jointly develop and use training programs with other companies
- Provide or access services provided by one company to a group of other companies, such as an online banking application managed by one company on behalf of affiliated banks
- Share news of common interest exclusively

Disadvantages of Extranet

- Extranets can be expensive to implement and maintain within an organization (e.g.: hardware, software, employee training costs) — if hosted internally instead of via an ASP
- Security of extranets can be a big concern when dealing with valuable information. System access needs to be carefully controlled to avoid sensitive information falling into the wrong hands.
- Extranets can reduce personal contact (face-to-face meetings) with customers and business partners. This could cause a lack of connections made between people and a company, which hurts the business when it comes to loyalty of its business partners and customers.

Data Transmission

Data transmission is essentially the same thing as Digital Communications, and implies physical transmission of a message as a digital bit stream, represented as an electromagnetic signal, over a physical point-to-point or point-to-multipoint communication channel. Examples of such channels are copper wires, optical fibers, wireless communication channels, and storage media. Data transmission is a subset of the field of data communications, which also includes computer networking or computer communication applications and networking protocols, for example routing and switching.

Data Transmission Rate and Bandwidth

Data Transmission Rate: The data transfer rate (DTR) is the amount of digital data that is moved from one place to another in a given time. The data transfer rate can be viewed as the speed of travel of a given amount of data from one place to another. In

telecommunications, data transfer is usually measured in bits per second. A kilobit is a thousand (10^3) bits. It's commonly used for measuring the amount of data that is transferred in a second between two telecommunication points. A kilobit per second is usually shortened to Kbps.

Data Transmission Bandwidth: Bandwidth is defined as the amount of data that can be transmitted in a fixed amount of time. For digital devices, the bandwidth is usually expressed in bits per second (bps) or bytes per second. For analog devices, the bandwidth is expressed in cycles per second, or Hertz (Hz).

A modem that works at 57,600 bps has twice the bandwidth of a modem that works at 28,800 bps.

In general, a link with a high bandwidth is one that may be able to carry enough information to sustain the succession of images in a video presentation. The greater the bandwidth of a given path, the higher the data transfer rate. It should be remembered that a real communications path usually consists of a succession of links, each with its own bandwidth. If one of these is much slower than the rest, it is said to be a *bandwidth bottleneck*.

Data Transmission Media

In order to carry data packets from one node to another on a network some sort of media must be employed. Transmission media is the path in which electrical signal travels from one place to other. Internet is a combination of different transmission media. The most common types of media are:

- Twisted Pair Copper Cable
- Coaxial Copper Cable
- Fibre Optic Cable or the wireless

FACTORS TO CONSIDER: Choosing the type of media is based on the protocol, topology of the network, distance, environment, security and speed. The speed of the internet also depends on the type of the media used for the internet access.

Twisted Pair Copper Cable:

Twisted pair cables have been used for connecting telephones to Local Exchanges ever since telephones were invented, the size and thickness have changed but they are basically the same as they were in the early twentieth century. A twisted pair cable consists of two insulated wires which are twisted round each other to form the twisted pair. The frequency of the twist down the cable coincides with the frequency of the electrical signals which will try to invade the cable from outside.

The twist will introduce a cancelling effect at the frequency of these signals thus reducing Radio Frequency Interference (RFI) and Electro-Magnetic Interference (EMI). RFI and EMI are introduced on the cable because the copper wire would act as an efficient aerial at certain frequencies if it were two straight cables. The base band signals being carried by the cable have sharp edges which can be severely distorted by the RFI, EMI, capacitance, resistance and inductance. Rounding the edges causes the receiving circuitry to think a "1" is a "0" and vice versa. Two main types of twisted pair cable exist –

- *Unshielded Twisted Pair (UTP) and*
- *Shielded Twisted Pair (STP)*

Unshielded Twisted Pair: The number of actual twisted pair cables held in one sheath varies, that used for connecting networks is generally 4 pairs (8 wires) but the cables laid in the road to feed telephones may have hundreds or thousands of pairs of cables. Each cable is identified by the main colour of the cable and a tracing colour. When there are hundreds of cables in a sheath there may be several tracer colours.

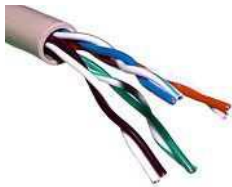
The type of cable used for networking have their pairs as the same colour but one will have a white cable with the same tracer colour as the main cable. Such as:

Orange and white with an orange tracer

Blue and white with a blue tracer

Green and white with a green tracer

Brown and white with a Brown tracer



Unshielded Twisted Pair (UTP) Cable

Shielded Twisted Pair (STP) has an extra metallic wrapper around the inner cables which will further reduce RFI and EMI, this sometimes surrounds all the pairs or could surround each individual pair this gives more bulk to the cable. Twisted pair cables are often shielded in attempt to prevent electromagnetic interference. Since the shielding is made of metal, it may also serve as a ground.

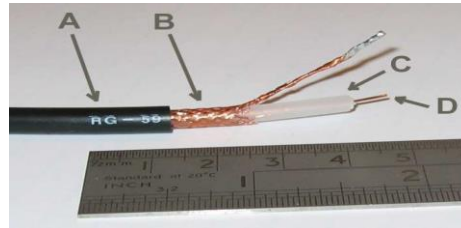
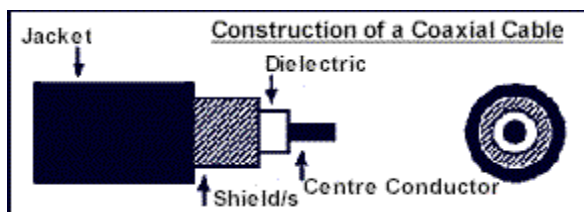


Shielded Twisted Pair

Coaxial Copper Cable:

The first networks to be designed were cabled with Coaxial cable. It has two wires like twisted pair but the design with an inner conductor and outer braid conductor enable it to

have a higher Bandwidth and be capable of higher data rates. Coaxial cable (or "coax") is the most common cable used for transmitting video signals. The name "coaxial" refers to the common axis of the two conductors.



Coaxial Cables

A coaxial cable has a solid copper or copper-clad-steel centre conductor surrounded by a non-conductive dielectric insulating material. The dielectric is surrounded by foil shield/s and/or copper braid/s which form the outer conductor and also shield against electromagnetic interference (EMI). The outer conductor/shield is encased in a PVC jacket.

The Advantages of Coaxial Cable

Broadband system: Coax has a sufficient frequency range to support multiple channels, which allows for much greater throughput.

Greater channel capacity: Each of the multiple channels offers substantial capacity. The capacity depends on where you are in the world. In the North American system, each channel in the cable TV system is 6MHz wide, according to the National Television Systems Committee (NTSC) standard. In Europe, with the Phase Alternate Line (PAL) standard, the channels are 8MHz wide. Within one of these channels, you can provide high-speed Internet access - that's how cable modems operate. But that one channel is now being shared by everyone using that coax from that neighbourhood node, which can range from 200 to 2,000 homes.

Greater bandwidth: Compared to twisted-pair, coax provides greater bandwidth system wide, and it also offers greater bandwidth for each channel. Because it has greater bandwidth per channel, it supports a mixed range of services. Voice, data, and even video and multimedia can benefit from the enhanced capacity.

Lower error rates: Since the inner conductor is in a Faraday shield, noise immunity is improved, and coax has lower error rates and therefore slightly better performance than twisted-pair. The error rate is generally 10^{-9} (i.e., 1 in 1 billion) bps.

Greater spacing between amplifiers: Coax's cable shielding reduces noise and crosstalk, which means amplifiers can be spaced farther apart than with twisted-pair.

The Disadvantages of coaxial Cables

Problems with the deployment architecture: The bus topology in which coax is deployed is susceptible to congestion, noise, and security risks.

Bidirectional upgrade required: In countries that have a history of cable TV, the cable systems were designed for broadcasting, not for interactive communications. Before they can offer to the subscriber any form of two-way services, those networks have to be upgraded to bidirectional systems.

Great noise: The return path has some noise problems, and the end equipment requires added intelligence to take care of error control.

High installation costs: Installation costs in the local environment are high.

Susceptible to damage from lightning strikes: Coax may be damaged by lightning strikes. People who live in an area with a lot of lightning strikes must be wary because if that lightning is conducted by a coax, it could very well fry the equipment at the end.

Fibre Optic Cables:

In recent years Fibre Optic Cables have been used increasingly as a Transmission media for data. Fibre Optic Cables are very thin (as thin as a human hair) strands of glass or plastic through which a laser light is passed from one end to the other. The strands are measured in μm (millionths of a metre). The thickness varies from 5 to $10\mu\text{m}$ for Single mode to 50 to $100\mu\text{m}$ for Multimode cables. The cables can be purchased with just one single fibre optic strand inside or with multiple strands.

Using Fibre Optic cable is very expensive due to its handling and the precision needed to attach the connector. Special tools, costing a lot of money, are used to attach the connector to the fibre strand. In recent years it has become apparent that fiber-optics are steadily replacing copper wire as an appropriate means of communication signal transmission. They span the long distances between local phone systems as well as providing the backbone for many network systems.

Other system users include cable television services, university campuses, office buildings, industrial plants, and electric utility companies. A fiber-optic system is similar to the copper wire system that fiber-optics is replacing. The difference is that fiber-optics use light pulses to transmit information down fiber lines instead of using electronic pulses to transmit information down copper lines.

Advantages of Fiber Optic Cable

Speed: Fiber optic networks operate at high speeds - up into the gigabits

Bandwidth: large carrying capacity

Distance: Signals can be transmitted further without needing to be "refreshed" or strengthened.

Resistance: Greater resistance to electromagnetic noise such as radios, motors or other nearby cables. **Maintenance:** Fiber optic cables costs much less to maintain.

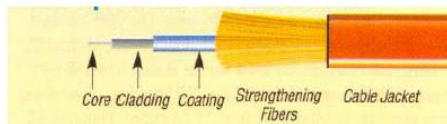
Fiber optic cables are much thinner and lighter than metal wires.

Data can be transmitted digitally (the natural form for computer data) rather than analogically.

Disadvantages of Fiber Optics

The cables are expensive to install.

The cables are more fragile than wire and are difficult to splice (link or join together).



Fibre Optics Cable

Classification of Computer Networks

Network Classification

The major categories used for classifying networks are geographical coverage, internetwork connection method and functional relationship (network architectures)

Classification of Networks by Internetwork

When two or more networks or network segments connected using devices that operate at layer 3 (the 'network' layer) of the OSI Basic Reference Model, such as a router is called internetwork. Any interconnection among or between public, private, commercial, industrial, or governmental networks may also be defined as an internetwork. In modern practice, the interconnected networks use the Internet Protocol. There are at least three variants of internetwork, depending on who administers and who participates in them:

- Intranet
- Extranet
- Internet

Classification of networks by Connection method

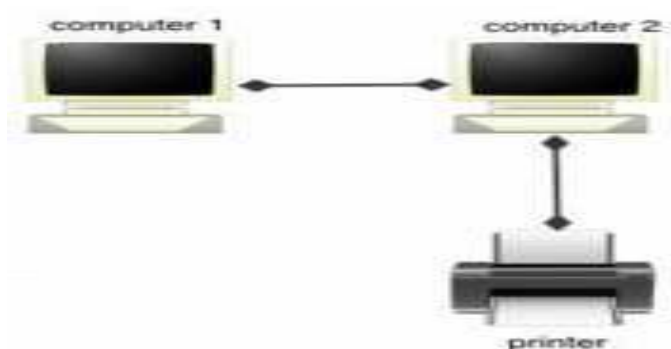
Computer networks can also be classified according to the hardware technology that is used to connect the individual devices in the network such as Optical fibre, Ethernet, Wireless LAN, HomePNA, or Power line communication. Ethernet uses physical wiring to connect devices. Often deployed devices are hubs, switches, bridges, and/or routers. Wireless LAN technology is designed to connect devices without wiring. These devices use radio waves as transmission medium.

Classification of Networks by Functional relationship (network architectures)

Computer networks may be classified according to the functional relationships which exist between the elements of the network, e.g., Active Networking, Client-server and Peer-to-peer (workgroup) architecture.

Peer-to-Peer: Peer-to-Peer networks (P2P) are the simplest of all LAN set-ups. Not only are they easy to configure and require few resources, they are also inexpensive to implement. Peer-to-Peer, also called P2P networks, are the most direct method of connecting computers, requiring only a network cable for the simplest of networks

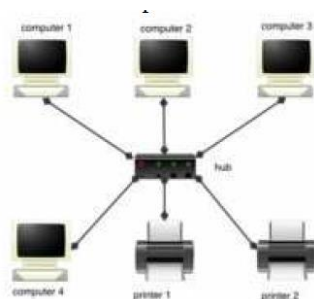
A simple Peer-to-Peer network can be a simple collection of two computers and a printer. Computer 1 will be able to print on the printer connected to Computer 2, if Computer 2 sets up the sharing permissions for the printer. Already we can see the advantages of building a small LAN: Files can be shared between the two computers and there is only one printer that must be purchased and maintained. A LAN like this is very simple to set up. In fact, many households may have LANs just like this.



Peer-to-Peer Network

When a LAN needs to include a larger number of computers, a hub may be used. This hub will enable all the computers connected to the network to communicate with each other without having to connect each individual computer to each and every other computer on the network. This saves space and considerable set-up time and expense.

Again, this P2P network is fairly simple to setup and allows all the computers to communicate with each other and to print using one or two printers. One note of caution - Printer 1 and Printer 2 in this diagram must be network enabled. If they are not, one could simply connect Printer 1 to Computer 4 and connect Printer 2 to Computer 3 and the printers could be shared as shown above.



Peer-to-Peer Network

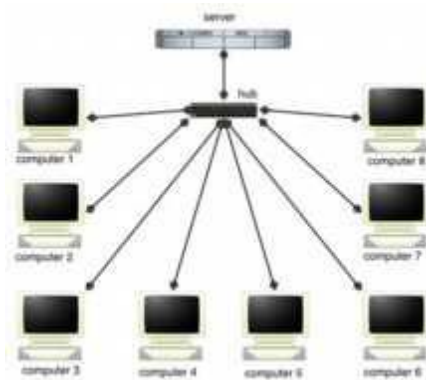
Advantages of a peer-to-peer Network

- Less initial expense - No need for a dedicated server.
- Setup - An operating system (such as Windows XP) already in place may only need to be reconfigured for peer-to-peer operations.

Disadvantages of a peer-to-peer Network

- Decentralized - No central repository for files and applications.
- Security - Does not provide the security available on a client/server network.

Client/Server: Client/Server LANs are generally much larger in scope than P2P networks. A file server is the most common server used. It frees up hard drive space on all the client machines and provides a single place to store files for everyone to access. Other popular servers include mail servers to send and deliver email, web servers to host Internet sites, database servers, and print servers. The complexity of client/server LANs can range from fairly simple networks to networks comprised of thousands of client machines and numerous servers that require an entire staff dedicated to keeping the network up and running.



Client/Server Network

Advantages of a client/server Network

- Centralized - Resources and data security are controlled through the server.
- Scalability - Any or all elements can be replaced individually as needs increase.
- Flexibility - New technology can be easily integrated into system.
- Interoperability - All components (client/network/server) work together.
- Accessibility - Server can be accessed remotely and across multiple platforms.

Disadvantages of a client/server network

- Expense - Requires initial investment in dedicated server.
- Maintenance - Large networks will require a staff to ensure efficient operation.
- Dependence - When server goes down, operations will cease across the network

Classification of Networks by Geographical Coverage or scale

Computer networks may be classified according to scale as follows:

Personal Area Network (PAN): A Personal Area Network (PAN) is a computer network used for communication among computer devices close to one person. Some examples of devices that are used in a PAN are printers, fax machines, telephones, PDAs or scanners. The reach of a PAN is typically within about 20-30 feet (approximately 6-9 metres).

Personal area networks may be wired with computer buses such as [USB](#) and FireWire. A Wireless Personal Area Network (WPAN) can also be made possible with network technologies such as IrDA and Bluetooth.

Local Area Network (LAN): A network covering a small geographical area, like a home, office, or building is a LAN. Current LANs are most likely to be based on Ethernet technology. For example, a library may have a wired or wireless LAN for users to interconnect local devices (e.g., printers and servers) and to connect to the internet.

Campus Area Network (CAN): A network that connects two or more LANs but that is limited to a specific and contiguous geographical area such as a college campus, industrial complex, or a military base. A CAN may be considered a type of MAN (metropolitan area network), but is generally limited to an area that is smaller than a typical MAN. This term is most often used to discuss the implementation of networks for a contiguous area. This should not be confused with a Controller Area Network. A LAN connects network devices over a relatively short distance. A networked office building, school, or home usually contains a single LAN, though sometimes one building will contain a few small LANs (perhaps one per room), and occasionally a LAN will span a group of nearby buildings. In TCP/IP networking, a LAN is often but not always implemented as a single IP subnet.

Metropolitan Area Network (MAN): A Metropolitan Area Network is a network that connects two or more Local Area Networks or Campus Area Networks together but does not extend beyond the boundaries of the immediate town/city. Routers, switches and hubs are connected to create a Metropolitan Area Network.

Wide Area Network (WAN): A WAN is a data communications network that covers a relatively broad geographic area (i.e. one city to another and one country to another country) and that often uses transmission facilities provided by common carriers, such as telephone companies. WAN technologies generally function at the lower three layers of the OSI reference model: the physical layer, the data link layer, and the network layer.

Global Area Network (GAN): Global Area Networks (GAN) specifications are in development by several groups, and there is no common definition. In general, however, a GAN is a model for supporting mobile communications across an arbitrary number of wireless LANs, satellite coverage areas, etc. The key challenge in mobile communications is "handing off" the user communications from one local coverage area to the next.

Principles of Networking

Computer Networks

Networks are systems that are formed by links. For example, roads that connect groups of people together create a physical network. Connections with your

friends create your personal network. Websites that allow individuals to link to each other's pages are called social networking sites.

People use the following networks every day:

- ☐ Mail delivery system
- ☐ Telephone system
- ☐ Public transportation system
- ☐ Corporate computer network
- ☐ The Internet

Networks share information and use various methods to direct the manner in which information flows. The information on the network goes from one place to another, sometimes via different paths, to arrive at the appropriate destination. For example, the public transportation system is a network similar to a computer network. The cars, trucks, and other vehicles are like the messages that travel within the network. Each driver defines a starting point (source computer) and an ending point (destination computer). Within this system, there are rules such as stop signs and traffic lights that control the flow from the source to the destination. A computer network also uses rules to control the flow of data between hosts on a network.

A host is any device that sends and receives information on the network. Some devices can serve either as hosts or peripherals. For example, a printer connected to your laptop that is on a network is acting as a peripheral. If the printer is connected directly to a network, it is acting as a host.

Many different types of devices can connect to a network:

- ☐ Desktop computers
- ☐ Laptop computers
- ☐ Tablet computers
- ☐ Smartphones
- ☐ Printers
- ☐ File and print servers
- ☐ Game consoles
- ☐ Home appliances

Computer networks are used globally in businesses, homes, schools, and government agencies. Many of the networks are connected to each other through the Internet. A network can share many different types of resources and data:

- Services, such as printing or scanning
- Storage space on removable devices, such as hard drives or optical drives □
- Applications, such as databases
- Information, stored on other computers □
- Documents
- Calendars, synchronizing between a computer and a smartphone

Logical Topologies

A logical topology describes how the hosts access the medium and communicate on the network. The two most common types of logical topologies are broadcast and token passing. In a broadcast topology, a host broadcasts a message to all hosts on the same network segment. There is no order that hosts must follow to transmit data. Messages are sent on a First In, First Out (FIFO) basis.

Token passing controls network access by passing an electronic token sequentially to each host. If a host wants to transmit data, the host adds the data and a destination address to the token, which is a specially-formatted frame. The token then travels to another host with the destination address. The destination host takes the data out of the frame. If a host has no data to send, the token is passed to another host.

Physical Topologies

A physical topology defines the way in which computers, printers, and other devices are connected to a network. The figure provides six physical topologies.

Bus Topology

In a bus topology, each computer connects to a common cable. The cable connects one computer to the next, like a bus line going through a city. The cable has a small cap installed at the end called a terminator. The terminator prevents signals from bouncing back and causing network errors.



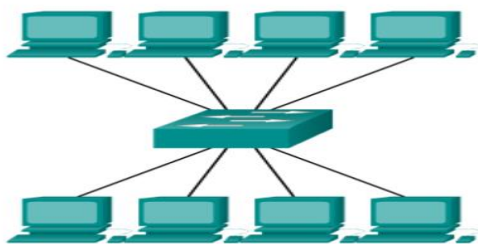
Ring Topology

In a ring topology, hosts are connected in a physical ring or circle. Because the ring topology has no beginning or end, the cable is not terminated. A token travels around the ring stopping at each host. If a host wants to transmit data, the host adds the data and the destination address to the token. The token continues around the ring until it stops at the host with the destination address. The destination host takes the data out of the token.



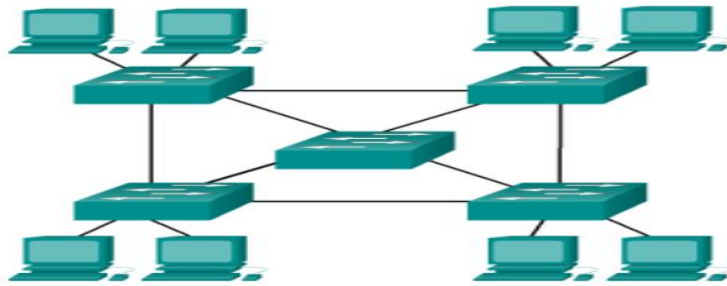
Star Topology

The star topology has a central connection point, which is normally a device such as a hub, switch, or router. Each host on a network has a cable segment that attaches the host directly to the central connection point. The advantage of a star topology is that it is easy to troubleshoot. Each host is connected to the central device with its own wire. If there is a problem with that cable, only that host is affected. The rest of the network remains operational.



Mesh Topology

The mesh topology connects all devices to each other. When every device is connected to every other device, a failure of any cable or device along a connection does not affect the network. The mesh topology is used in WANs that interconnect LANs.



Internet Connectivity

Introduction

There are several methods that can be used to connect to the Internet. Connection speed has become a vital factor as the requirement to download and upload large files becomes increasingly important. This is something that should be considered when reviewing the options for connecting to the Internet.

Internet Connection Methods

If you want to connect to the Internet, here are the options.

- **Wireless:** Choose this if you have a wireless router or network, or if you're connecting to a hotspot, even if you also have a broadband connection.
- **Broadband:** Choose this if your computer is connected directly to a broadband modem (also called a Digital Subscriber Line [DSL] or cable modem), and you have a Point-to-Point Protocol over Ethernet (PPPoE) Internet account. With this type of account, you need to provide a user name and password to connect.
- **Dial-up:** Choose this if you have a modem but it is not a DSL or cable modem, or if you want to use Integrated Services Digital Network (ISDN) to connect your computer to the Internet.

Basic hardware requirements for Internet connectivity

To connect to the Internet you will need:

- **A computer plus a modem** (or access to an Internet connected Local Area Network).
- **Access to either a phone line** (or in the case of large organizations, a Local Area Network, which in turn allows connection to the Internet).
- **You will need an Internet account.** In the case of an individual computer connected to the Internet via a phone line, this will normally come as a package allowing you to

access the Internet via an ISP (Internet Service Provider) and usually also includes an email account, which allows you to send and receive email via the Internet.

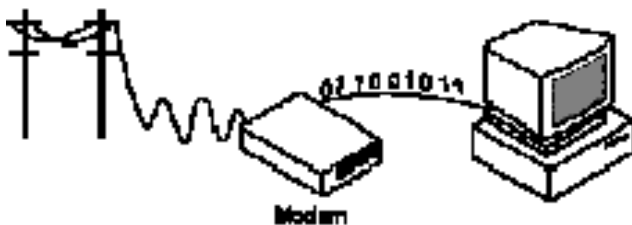
Lastly you will require an Internet browser (i.e. Microsoft Internet Explorer), which is software that allows you to view Web pages.

What is a MODEM?

Modem (from **MO**dulator-**DE**Modulator) is a device that modulates an analog carrier signal to encode digital information, and also demodulates such a carrier signal to decode the transmitted information. The goal is to produce a signal that can be transmitted easily and decoded to reproduce the original digital data.

A modem is a device or program that enables a computer to transmit data over, for example, telephone or cable lines. Computer information is stored digitally, whereas information transmitted over telephone lines is transmitted in the form of analog waves. A modem converts between these two forms.

Modems are generally classified by the amount of data they can send in a given time, normally measured in bits per second, or "bps". They can also be classified by Baud, the number of times the modem changes its signal state per second. Faster modems are used by Internet users every day, notably cable modems and ADSL modems.



Types of Modems

- Internal Modem
- External Modem

There is one standard interface for connecting external modems to computers called *RS-232*; any external modem can be attached to any computer that has an RS-232 port, which almost all personal computers have. There are also modems that come as an expansion board that you can insert into a vacant expansion slot. These are sometimes called *onboard* or *internal modems*.



External Modem



Internal Modems

While the modem interfaces are standardized, there are a number of different protocols for formatting data to be transmitted. Aside from the transmission protocols that they support, the following characteristics distinguish one modem from another:

- **Bps:** How fast the modem can transmit and receive data. At slow rates, modems are measured in terms of baud rates. The slowest rate is 300 baud (about 25 cps). At higher speeds, modems are measured in terms of bits per second (bps). The fastest modems run at 57,600 bps, although they can achieve even higher data transfer rates by compressing the data. Obviously, the faster the transmission rate, the faster you can send and receive data.
- **Voice/data:** Many modems support a switch to change between voice and data modes. In data mode, the modem acts like a regular modem. In voice mode, the modem acts like a regular telephone. Modems that support a voice/data switch have a built-in loudspeaker and microphone for voice communication.
- **Auto-answer:** An auto-answer modem enables your computer to receive calls in your absence. This is only necessary if you are offering some type of computer service that people can call in to use.
- **Data compression:** Some modems perform data compression, which enables them to send data at faster rates. However, the modem at the receiving end must be able to decompress the data using the same compression technique.
- **Flash memory:** Some modems come with *flash memory* rather than conventional ROM, which means that the communications protocols can be easily updated if necessary.
- **Fax capability:** Most modern modems are fax modems, which means that they can send and receive

Connecting to the Internet

Setting up an Internet Connection

Step 1: Getting Started

To connect to the Internet, you need a PPP or SLIP account with an Internet Service Provider (ISP). You also need the following information from your ISP:

- User name
- Password
- Local access phone number or dial-up number
- Your host and domain name
- DNS server IP address
- Authentication technique (whether or not a terminal window is used)

Step 2: Create a New Connection

- From the **Start** menu, Click on **Control Panel**
- Click on **Network and Internet Connections**
- Select **Set up or change your Internet connection**
- Click on **Setup**

Step 3: New Connection Wizard

- When the New Connection Wizard starts up, click the **Next** button.
- Select “**Connect to the Internet**” and click the **Next** button.
- Choose the option to “**manually setup the connection**” and click the **Next** button.
- Choose the option to connect using a “**Dial-Up Modem**” and click the **Next** button.
- You should now be prompted to name your internet connection. Enter the ISP name and click the **Next** button.
- For the phone number, find the appropriate number and enter it into the Phone Number field:
- On the **Internet Account Information** dialog box, enter your *username* and *password*.
- Check the 1st box if you want everyone or just yourself to be able to use this connection,
- Check the 2nd box if you want to make this connection your default and click the **Next** button.

- When the Completing the New Connection Wizard window shows up, you can check the box to add a shortcut to this connection on your desktop, click the **Finish** button.

Connecting to the Internet via wireless or Wi-Fi

Wi-Fi (Wireless Fidelity) is a way to connect to the Internet without wires or cables. When you use Wi-Fi technology, you connect wirelessly to a base station that provides the link to the Internet or other type of network.

One of the benefits of Wi-Fi technology is its ability to quickly and wirelessly connect your handheld or laptop to the Internet and your corporate network when you are away from the office. Many cafes, coffee shops, libraries, airports, and other public places offer Wi-Fi Access Points for general use by the public. These public Wi-Fi Access Points are often called **HotSpots**. Using a Wi-Fi connected handheld, you can connect to various public Hot Spots.

Things you'll need:

- Computer with wireless capability
- Wireless router or hot spot

Step 1

Some computers connect automatically. The only thing that you have to do is make sure the wireless in your computer is turned on. There should be a button with an indicator light near your keyboard.

Step 2

For older computers more steps are needed. With a computer running windows open your start menu and find “network” or “internet connections”. There should be an option for wireless that you can click. Your computer will search for wireless connections and you will be offered a list of options. Find the connection in the list for the location that you are at. Once you click it you will be connected.

Step 3

For wireless connections that have passwords the wireless wizard is your best option. It will allow you to first enter the SSID which is the name of the network. Usually the SSID and password are posted in locations where the access is free. The SSID is the name of the network. Keep clicking next through the screens and enter the user name and password in the indicated fields.

Choices of Internet Access

Depending on your location, you might have several different choices for Internet access. Starting with the most attractive technologies for home Internet access, common Internet connection types are:

- **Cable modems:** The best performing and most affordable option available to customers, most cable TV providers offer broadband Internet access.
- **DSL:** Digital Subscriber Line. An excellent choice for businesses, DSL typically offers better reliability than cable modems. However, DSL tends to be more expensive than cable modems for similar levels of service.
- **Dial-up:** The slowest method of connecting to the Internet, dial-up enables you to connect to the Internet using your existing phone lines. Dial-up is convenient because it is available to any location with a phone. However, slow performance makes using the Internet frustrating.
- **Satellite:** Satellite broadband services provide high-speed Internet access to any location with a clear view of the sky (currently available in North America and certain other locations). Satellite services may be the only broadband option for people living in rural areas. The cost of satellite services is significantly higher than other services. While you can transfer large files quickly with satellite, browsing the Web or playing online games can seem slower than with dial-up because of the delay caused by sending signals to and from satellites.

What is broadband?

Broadband is the generic term that is used to cover the new generation of Internet connections. There is no one precise definition as the term is used to describe a range of services, but they all have common key features such as:

- fast access speeds
- a permanent "always-on" connection to the Internet
- ability to handle large files quickly
- flat-rate monthly charges

There are a number of different types of broadband - it is available by telephone line, cable, satellite and wireless networks, among others. The different types have varying speeds and costs. Where you live will determine which options are available to you.

Wireless Broadband and Bandwidth

Internet bandwidth (the speed at which your computer can send and receive information) is measured in either Kbps (kilobits per second) or Mbps (megabits per second). If you are lucky enough to have multiple broadband options in your area, compare these factors:

- **Downstream bandwidth.** This is the speed with which your computer can receive information from the Internet. The higher the downstream bandwidth, the faster your computer can display Web pages, transfer music, and download files. For most people, downstream bandwidth is more important than upstream bandwidth, so the speeds tend to be much higher. For example, a cable modem service might offer 6,000 Kbps downstream and only 768 Kbps upstream.
- **Upstream bandwidth.** This is the amount of data your computer can send to the Internet. This isn't important if you just plan to read e-mail and surf the Web, because your computer only needs to send a small request in order to receive a large Web page or email. However, if you're into online gaming or you want to send large files to people, then higher upstream bandwidth is important, and you should choose the highest upstream bandwidth available.

Wireless broadband

Wireless is a collective term used to describe different technologies that use radio transmitters and receivers to link computers. Wireless services are quickly gaining popularity across the world and, as they do, products and services are being developed and upgraded to meet demand. For example, where the adoption of broadband has been limited due to lack of broadband-enabled telephone exchanges, or businesses being too remote, broadband has been delivered by using variations of wireless technology.

WiFi

Wireless local area network hotspots, known as WiFi hotspots, are being rolled out globally. They can be subscription only, pay-per-use or free to use and they often appear in airports, hotels, cafes, pubs and motorway service stations. Some operators also offer mobile phone style "roaming" facilities.

Wimax

Wimax (Worldwide Interoperability for Microwave Access) is a fast-emerging alternative to WiFi. Wimax offers a much greater reach than the existing WiFi broadband, making it possible to provide broadband to areas where other technology is unsuitable.

Advantages of wireless broadband

- High bandwidth.
- Access the web on the move.
- High media profile makes advice easy to come by.

Disadvantages of wireless broadband

- Still in its infancy, so availability is limited.
- Setting up a wireless network requires substantial technical knowledge.
- Some security concerns persist about wireless technology.

Wireless Transmission Media

Introduction

Wireless networking simply refers to creating a network with no wires or cables. This is achieved in one of 2 ways, either using radio frequency or an infra-red port connection. Wireless internet is a new technology and not many internet service providers offer it. Wireless internet refers to establishing an internet connection over a wireless network via an antenna. Wireless Internet connections do not require a modem.

Benefits of Wireless Internet connection

A wireless internet connection is very fast and stable as opposed to other broadband solutions. Wireless internet does not need to be connected to any main phone line hence the line would be freed for other uses. To connect to wireless internet would just be a matter of setting up a long range antenna which will establish a connection as soon as it can pick up our signal.

Wireless internet is also very affordable and is also very easy to install. It is just a matter of installing all the hardware and software and you'll be ready to access the internet. Wireless networks are also very efficient. In some instances Wireless Internet has proven to be faster than a broadband connection. Most broadband solutions offer a fixed download/upload ratio, but with wireless, the speed at which you download can be the same as what you upload.

Hence you are not restricted to fast download speeds and slow upload speeds. This makes wireless versatile and a worthwhile solution for businesses in demand of fast internet access.

Wireless Internet connection options

VSAT (Very Small Aperture Terminal) Internet connection

A VSAT is a small-sized telecommunications earth station that transmits and receives via satellite. The terminal size is 1.2 to 2.4 meter in diameter. VSATs are becoming increasingly popular, because they are a single, flexible communications platform that can be installed quickly and cost effectively to provide telecoms solutions for consumers, governments and corporations.

They have been in use for more than 10 years and, with more than 500,000 systems operating in more than 120 countries, VSATs are a mature and proven technology. VSAT networks may transmit and receive voice, data, fax, or video conferencing. VSAT represents a cost-effective solution for those who want an independent communications network to connect sites, especially sites where any other connectivity options are not possible or realistic to implement.

VSAT Applications

VSAT networks offer value-added satellite-based services capable of supporting the Internet, data, LAN, voice/fax communications, and can provide powerful, dependable, private and public network communications solutions. VSATs are used for a wide variety of telecommunications applications, including corporate networks, Internet service provision, rural telecoms, distance learning, telemedicine, cybercafés, banking, oil and gas exploration, manufacturing, government agencies and departments and much more.

As Internet traffic continues to grow at exponential rates worldwide, Internet services providers (ISP's) everywhere are turning to satellite to keep with demand for network bandwidth. VSAT has offered ISP's and telecommunications service providers' easily scalable, cost-effective Internet access since 1989. A VSAT has certain clear advantages over terrestrial networks and other wireless communications media.

Fast & smooth installation: Satellite services are usually activated much more quickly than terrestrial networks. Since meters or kilometers of cable need not be installed. An antenna, modem and satellite circuit can be installed and made operational in just a few weeks rather than the months installation of cable may take.

It is easily implemented due to the wireless nature of connectivity. And as a network grows, additional capacity can be obtained in just as short a time.

Satellite networks offer much higher reliability than terrestrial networks. Network performance is enhanced with satellite by linking directly to an Internet backbone, bypassing congested terrestrial lines, exchanges and numerous interconnections. This reduces the potential points of failure.

Economic Network Costs

The broadcast nature of satellites allows for the simultaneous delivery of information to wide geographic areas without regard to terrestrial infrastructure or geographic barriers. Charges for terrestrial services are nearly always distance-dependent, while VSAT connections cost the same whether sites are one or 1000 Km apart. VSATs are capable of sending and receiving all sorts of video, data and audio content at the same high speed regardless of their distance from terrestrial switching offices and infrastructure.

Telephony and broadband infrastructure require heavy infrastructure and are particularly expensive to deploy and maintain especially to remote and unserved locations. And with most VSAT services the cost-per-connection comes down considerably when a customer adds users. Technology is also changing rapidly to make satellite access to the Internet more affordable.

Global Availability

The key advantage of a VSAT earth station, versus a typical terrestrial network connection, is that the reach of buried cable does not limit VSATs. Satellite is the only telephony and broadband wide-area network technology that is available everywhere - in even the most remote urban and rural areas, rain forests or concrete jungles, anywhere in the world. A VSAT earth station can be placed anywhere - as long as it has an unobstructed view of the satellite.

Site Relocation and Addition

Relocating and adding network sites is much simpler and cheaper with a satellite network than with most terrestrial technologies.

Flexibility & Expansion

VSAT networks are highly scalable, i.e. it provides the ability to scale according to the customer's requirements with low incremental cost. This is especially important as businesses grow over time. A VSAT network can support thousands of VSAT remotes. Hence, a customer may start with Data services (internet browsing & e-mail) and then adding more bandwidth can

be done over time by a simple reconfiguration at the Network Operating Center. In addition customers are able to add voice capabilities or video applications when the need arises.

VSAT Bands

There are 3 major signal bands available on the VSAT system. They are the *C-band*, *KUband* and the *KA- band*.

The **C-Band** is the lowest frequency signal based transponder, which exhibits reliable capability to withstand weather interference. It also allows the largest bandwidth upgradability.

The **KU band** is subject to weather change interference. A prolonged rain or cloudy weather sometimes affects transmission.

Typically, interactive Ku-band antenna sizes range from 75 centimetres to 1.8 metres and Cband from 1.8 metres to 2.4 metres.

Wireless Radio Frequency Internet Connection

Wireless Internet is achieved by sending data on a radio frequency rather than the conventional way through copper wiring (phone lines). This is achieved by using a long range antenna and a wireless network card. The antenna will pick up radio waves from a Wireless Internet access point and send data to your server terminal through a Wireless Network Card.

This concept works similar to watching TV or listening to the radio. As soon as your hardware is up and running a connection would be established. Your computer would then be able to send and receive information on this network. For wireless Internet to be fully operational a line of sight must be visible between the antenna and our access point. Distance depends on how powerful the antenna is, but typically the closer an antenna is to an access point, the better the reception will be. Hence the transfer speed would be greater.

Wireless Internet Locations

To be able to establish a wireless internet connection, an antenna must be within 22kms and in the line of sight of an access point. You will need to contact your ISP and discuss your location, specifically whether you are in one of their coverage areas or not.

Costs of Wireless Internet

The cost for Wireless internet depends on the transfer speed of your connection and how much data you wish to download. A Setup fee consisting of an Access Point, Antenna, Mast, Mounting Kit, labour to install and configure Cable and sundries will also be paid.

Obstacles to Effective Transmission

If you are located more than 2 miles from a cell, the wireless Internet connection will require a direct radio line of sight. As a general rule, if you cannot get a direct radio line of sight from your location to an access point, you can't get wireless access.

- Busy lines while trying to establish phone connection with ISP
- Congestion of bandwidth leading to slowing down of speed
- Bad Weather conditions
- Outages from ISP during Maintenance
- Power failure

Network Protocol

Network Protocol

In computing, a protocol is a convention or standard that controls or enables the connection, communication, and data transfer between two computing endpoints. In its simplest form, a protocol can be defined as the rules governing the syntax, semantics, and synchronization of communication. Protocols may be implemented by hardware, software, or a combination of the two.

Typical properties of protocols

It is difficult to generalize about protocols because they vary so greatly in purpose and sophistication. Most protocols specify one or more of the following properties:

- Detection of the underlying physical connection (wired or wireless), or the existence of the other endpoint or node
- Handshaking:
- Negotiation of various connection characteristics
- How to start and end a message
- How to format a message
- What to do with corrupted or improperly formatted messages (error correction)
- How to detect unexpected loss of the connection, and what to do next
- Termination of the session and or connection.

Importance of Network protocols

The widespread use and expansion of communications protocols is both a prerequisite for the Internet, and a major contributor to its power and success. The pair of Internet Protocol (or IP)

and Transmission Control Protocol (or TCP) are the most important of these, and the term TCP/IP refers to a collection (or protocol suite) of its most used protocols. Most of the Internet's communication protocols are described in the RFC documents of the Internet Engineering Task Force (or IETF).

The need for protocols also applies to network devices. Computers have no way of learning protocols, so network engineers have written rules for communication that must be strictly followed for successful host-to-host communication. These rules apply to different layers of sophistication such as which physical connections to use, how hosts listen, how to interrupt, how to say good-bye, and in short how to communicate, what language to use and many others. These rules, or protocols, that work together to ensure successful communication are grouped into what is known as a protocol suite.

Types of network protocols

- IP (Internet Protocol)
- UDP (User Datagram Protocol)
- TCP (Transmission Control Protocol)
- DHCP (Dynamic Host Configuration Protocol)
- HTTP (Hypertext Transfer Protocol)
- FTP (File Transfer Protocol)
- Telnet (Telnet Remote Protocol)
- SSH (Secure Shell Remote Protocol)
- POP3 (Post Office Protocol 3)
- SMTP (Simple Mail Transfer Protocol)
- IMAP (Internet Message Access Protocol)

Advantages of TCP/IP for Internet Connectivity

The **Internet Protocol Suite** (commonly **TCP/IP**) is the set of communications protocols used for the Internet and other similar networks. It is named from two of the most important protocols in it: the Transmission Control Protocol (TCP) and the Internet Protocol (IP), which were the first two networking protocols defined in this standard. The Internet Protocol Suite, like many protocol suites, may be viewed as a set of layers. Each layer solves a set of problems involving the transmission of data, and provides a well defined service to the upper layer protocols based on using services from some lower layers. Upper layers are logically

closer to the user and deal with more abstract data, relying on lower layer protocols to translate data into forms that can eventually be physically transmitted.

The Internet Protocol (IP) is the method or protocol by which data is sent from one computer to another on the Internet. Each computer (known as a host) on the Internet has at least one IP address that uniquely identifies it from all other computers on the Internet. When you send or receive data (for example, an e-mail note or a Web page), the message gets divided into little chunks called packets. Each of these packets contains both the sender's Internet address and the receiver's address. Any packet is sent first to a gateway computer that understands a small part of the Internet.

The gateway computer reads the destination address and forwards the packet to an adjacent gateway that in turn reads the destination address and so forth across the Internet until one gateway recognizes the packet as belonging to a computer within its immediate neighbourhood or domain. That gateway then forwards the packet directly to the computer whose address is specified. Because a message is divided into a number of packets, each packet can, if necessary, be sent by a different route across the Internet. Packets can arrive in a different order than the order they were sent in. The Internet Protocol just delivers them. It's up to another protocol, the Transmission Control Protocol (TCP) to put them back in the right order.

IP is a connectionless protocol, which means that there is no continuing connection between the end points that are communicating. Each packet that travels through the Internet is treated as an independent unit of data without any relation to any other unit of data. (The reason the packets do get put in the right order is because of TCP, the connection-oriented protocol that keeps track of the packet sequence in a message.). In the Open Systems Interconnection (OSI) communication model, IP is in layer 3, the Networking Layer.

The most widely used version of IP today is Internet Protocol Version 4 (IPv4). However, IP Version 6 (IPv6) is also beginning to be supported. IPv6 provides for much longer addresses and therefore for the possibility of many more Internet users. IPv6 includes the capabilities of IPv4 and any server that can support IPv6 packets can also support IPv4 packets

Concept of Internet Service Provider (Isp)

Internet Service Provider (ISP)

An **Internet service provider (ISP)**, also called **Internet access provider** or **IAP**) is a company which primarily offers their customers access to the Internet using dial-up or other means of data telecommunication. ISPs may provide Internet e-mail accounts to users which allow them to communicate with one another by sending and receiving electronic messages through their ISPs' servers. ISPs may provide other services such as remotely storing data files on behalf of their customers, as well as other services unique to each particular ISP. The most important services an ISP offers are:

- **Internet access.** Access any Web site, send instant messages to your friends, play online games, or use any other Internet service.
- **E-mail.** You can access your e-mail with Microsoft Outlook Express or your Web browser. Most ISPs offer multiple e-mail addresses, so everyone in your family can have an account. ISPs typically provide spam filtering that reduces, but does not eliminate, unwanted messages.

Factors to be considered when choosing an ISP

Bandwidth: Internet bandwidth (the speed at which your computer can send and receive information) is measured in either Kbps (kilobits per second) or Mbps (megabits per second)

Reliability and Customer Service: ISP reliability has increased significantly in recent years; however, it is still not as reliable as your phone or television service. There is no objective way to measure reliability and customer service, so you should talk to your neighbours about their experiences and search the Web for reviews of ISPs in your area.

Technical Strength: Of major concern to you must be whether the ISP has the technical capabilities to meet your present and future Internet requirements. The ISP itself has to tell you what its technical strength is like, i.e. in terms of personnel and equipment. In view of the number of “cowboys” claiming to be ISPs, such information provided by the ISP should be verified. Always ask technical questions, no matter how awkward. Is onsite support available? What is their response time to issues like? Is emergency support available? What are the limits, if any, on disk storage, e-mail or other Internet traffic?

Busy Signals: Busy signals are an indication of performance problems that affect Internet usage. What is the ISP's policy on busy signals? How busy are their modems? When are the

peak usage hours? How will the ISP protect you from busy signals? Failure to demand for truthful answers can eventually turn out to be costly in terms of time, funds and opportunities.

Professionalism: The professionalism of the prospective ISP is equally important. How professional is the outfit's customer service? Are they the sort who promise heaven but deliver hell? You have to be careful not to be taken in by sugar coated stories, or window dressed adverts. Always separate the hype from the real facts on the ground. Are staffs courteous, trained and professional? Remember you're the customer. How are you treated when contacting the ISP? Was your call or request treated in a professional and timely manner? Or are you given the impression when you make enquiries that you are "disturbing, or asking too many questions"?

Total Internet Solutions: Additionally, assess the ISP's attitude to Total Internet Solutions (TIS). Any truly professional ISP should be able to provide TIS, not just an Internet connection. Can the ISP fulfil all your needs? Can it provide services such as Domain Name Services, Onsite installation and maintenance, Web services, Internet security services, and Network Consulting? It is important that you identify and choose an ISP who is focused on solutions - TIS. Providing an Internet connection is not enough.

Cost: The cost of the Internet service is of course important. But it must be stressed again that cost should not be the sole determining factor in choosing an ISP. Some ISPs might offer very cheap rates, as a way of attracting customers, while the quality of their service is very poor. In such a situation, you find that your Internet service is actually very expensive as the only Internet services you're paying for are excuses and busy signals.

Charges/ Payment Methods: In addition, how does the ISP charge for its services? Are charges based on flat rate access (which requires you to pay a fixed amount for a given period, independent of usage), or usage access – "pay-as-you-go". The charge / payment method used by the ISP must be one you are comfortable with, and must make financial sense to you. You must have a focus on how you want to use the Internet. Furthermore, are there any hidden or extra charges?

Track Record: Another crucial factor is the track record. What is the track record of the ISP? This could be track record of the ISP itself, as well as the track record of key professionals in the organization. With regard to the track record of the ISP, the ISP should be able identify successful working sites as evidence of past and current performance. The previous

performance of the key professionals is important, especially if it is a new ISP, with a few clients. To really assess the track record of the ISP, it is important that as a prospective client of the ISP, you meet with some clients of the ISP independently and ask for

The future: What is an ISP's plan for the future? This is important since the Internet is a medium for the future, and in view of the rapidity of developments on the Internet. Does the ISP have an idea of how it will offer quality services in future, not just today? Anything as dynamic as the Internet changes daily. Does it have any future plans, or it is just going along for the ride? What are the ISP's growth plans? If such plans exist are they realistic? Simply put, the ISP must have a vision of the future. The ISP's vision of the future depends to a large extent on its understanding of the future. There is no room for compromise. Go for an ISP that understands the future - of Internet. By doing this one is securing a fruitful, long-lasting and problem-free Internet experience. The ISP's vision of the future influences its internal training programme. A lot can be detected by observing its attitude towards training. Does the ISP invest in training? And if it does, is it practical, result-oriented, and in line with its vision for the future? Or is it simply to give the "techies" a break?

Domain Name System

What is a Domain Name?

In computer networking, a **domain name** is a name given to a collection of network devices that belong to a **domain** which is managed according to some common property of the members or within a common administrative boundary. In particular, the term is used to describe the regions of administrative authority within the Domain name system used for the Internet. It is a name that identifies one or more *IP addresses*. For example, the domain name *microsoft.com* represents about a dozen IP addresses. Domain names are used in URLs to identify particular Web pages. For example, in the URL

http://www.federalpolytechnicmubi.edu.ng/index.html, the domain name is *federalpolytechnicmubi.edu.ng*.

Every domain name has a suffix that indicates which Top Level Domain (TLD) it belongs to. There are only a limited number of such domains. For example:

- **gov** - Government agencies
- **edu** - Educational institutions
- **org** - Organizations (nonprofit)

- **mil** - Military
- **com** - commercial business
- **net** - Network organizations
- **ca** - Canada
- **th** - Thailand

The Internet is based on IP addresses, not domain names, every Web server requires a Domain Name System (DNS) server to translate domain names into IP addresses.

Uniform Resource Locator (URL)

URL is the global address of documents and other resources on the World Wide Web. The first part of the address is called a *protocol identifier* and it indicates what protocol to use and the second part is called a *resource name* and it specifies the IP address or the domain name where the resource is located. The protocol identifier and the resource name are separated by a colon and two forward slashes. For example, the two URLs below point to two different files at the domain *pcwebopedia.com*.

The first specifies an executable file that should be fetched using the FTP protocol; the second specifies a Web page that should be fetched using the HTTP protocol:

ftp://www.federalpolytechnicmubi.edu.ng/stuff.exe

http://federalpolytechnicmubi.edu.ng/index.html

IP Addresses

IP Address is an identifier for a computer or device on a TCP/IP network. An **IP address** (Internet Protocol Address) is a logical address of a network adapter. The **IP address** is unique and identifies computers on a network. An **IP address** can be private, for use on a LAN, or public, for use on the Internet or other WAN.

IP address can be changed and hidden. Networks using the TCP/IP protocol route messages based on the IP address of the destination. The format of an IP address is a 32-bit numeric address written as four numbers separated by periods. Each number can be zero to 255. For example, 1.160.10.240 could be an IP address. Within an isolated network, you can assign IP addresses at random as long as each one is unique. However, connecting a private network to the Internet requires using registered IP addresses (called Internet addresses) to avoid duplicates.

The four numbers in an IP address are used in different ways to identify a particular network and a host on that network. Four regional Internet registries -- ARIN, RIPE NCC, LACNIC and APNIC -- assign Internet addresses from the following three classes.

Class A - supports 16 million hosts on each of 126 networks

Class B - supports 65,000 hosts on each of 16,000 networks

Class C - supports 254 hosts on each of 2 million networks

The number of unassigned Internet addresses is running out, so a new classless scheme called CIDR is gradually replacing the system based on classes A, B, and C and is tied to adoption of IPv6.

Web Server

A web server is a computer that delivers (*serves up*) Web pages. Every Web server has an IP address and possibly a domain name. For example, if you enter the URL *http://www.pcwebopedia.com/index.html* in your browser, this sends a request to the server whose domain name is *pcwebopedia.com*. The server then fetches the page named *index.html* and sends it to your browser.

Any computer can be turned into a Web server by installing server software and connecting the machine to the Internet. There are many Web server software applications, including public domain software from NCSA and Apache, and commercial packages from Microsoft, Netscape and others.

How Web Servers Work

Have you ever wondered just exactly how a Web page you are reading found its way into your browser and onto your computer screen? The process largely depends on Web servers. Typically, users visit a Web site by either clicking on a hyperlink that brings them to that site or keying the site's URL directly into the address bar of a browser. But how does the same site appear on anyone's computer anywhere in the world and often on many computers at the same time?

Let's use Webopedia as an example. You decide to visit Webopedia by typing its URL -- *http://www.webopedia.com* -- into your Web browser. Through an Internet connection, your browser initiates a connection to the Web server that is storing the Webopedia files by first converting the domain name into an IP address (through a domain name service) and then

locating the server that is storing the information for that IP address (also see Understanding IP Addressing).

The Web server stores all of the files necessary to display Webopedia's pages on your computer -- typically all the individual pages that comprise the entirety of a Web site, any images/graphic files and any scripts that make dynamic elements of the site function.

Once contact has been made, the browser requests the data from the Web server, and using HTTP, the server delivers the data back to your browser. The browser in turn converts, or formats, the computer languages that the files are made up of into what you see displayed in your browser. In the same way the server can send the files to many client computers at the same time, allowing multiple clients to view the same page simultaneously.

Domain Name System (DNS)

The **Domain Name System** (DNS) is the way that Internet domain names are located and translated into Internet Protocol addresses. A domain name is a meaningful and easy-to-remember "handle" for an Internet address. Maintaining a central list of domain name/IP address correspondences would be impractical, the lists of domain names and IP addresses are distributed throughout the Internet in a hierarchy of authority. There is probably a DNS server within close geographic proximity to your access provider that maps the domain names in your Internet requests or forwards them to other servers in the Internet.

The Domain Name System (DNS) associates various information with domain names; most importantly, it serves as the "phone book" for the Internet by translating human-readable computer hostnames, e.g. *www.example.com*, into IP addresses, e.g. *208.77.188.166*, which networking equipment needs to deliver information. A DNS also stores other information such as the list of mail servers that accept email for a given domain. By providing a worldwide keyword-based redirection service, the Domain Name System is an essential component of contemporary Internet use.

The Domain Name Space

The domain name space consists of a tree of domain names. Each node or leaf in the tree has zero or more *resource records*, which hold information associated with the domain name. The tree sub-divides into *zones* beginning at the root zone. A DNS zone consists of a collection of connected nodes authoritatively served by an *authoritative DNS name server*. (Note that a single name server can host several zones.)

When a system administrator wants to let another administrator control a part of the domain name space within the first administrator's zone of authority, control can be delegated to the second administrator. This splits off a part of the old zone into a new zone, which comes under the authority of the second administrator's name servers. The old zone ceases to be authoritative for the new zone.

Parts of a Domain Name

A domain name usually consists of two or more parts (technically a **label**), which is conventionally written separated by dots, such as example.com. The rightmost label conveys the top-level domain (for example, the address www.example.com has the top-level domain com).

Each label to the left specifies a subdivision, or sub domain of the domain above it. Note: "sub domain" expresses relative dependence, not absolute dependence. For example: example.com is a sub domain of the com domain, and www.example.com is a sub domain of the domain example.com. In theory, this subdivision can go down 127 levels. Each label can contain up to 63 octets. The whole domain name may not exceed a total length of 253 octets. In practice, some domain registries may have shorter limits.

A hostname refers to a domain name that has one or more associated IP addresses; i.e. the 'www.example.com' and 'example.com' domains are both hostnames, however, the 'com' domain is not.

DNS Servers

The Domain Name System is maintained by a distributed database system, which uses the client-server model. The nodes of this database are the name servers. Each domain or sub domain has one or more authoritative DNS servers that publish information about that domain and the name servers of any domains subordinate to it. The top of the hierarchy is served by the root name servers

Obstacles to Internet Growth in Nigeria

Introduction

This study examined the relationship between the Internet development and various social, economic and political factors that are hypothesized to affect the Internet growth. This finding shows that the Internet penetration is related to a country's wealth, telecommunication infrastructure, urbanization and stability of the government, but not related to the literacy level,

political freedom and English proficiency.

Suggestions

THE advent of Internet, perhaps, has been considered as being as important in its effects on society as that of the telephone, personal computer or even the printing press. Undoubtedly, Internet today is the number one vehicle for commerce, business, communication, governance, manpower development, research among others.

If Nigeria is to really live up to its aspirations of emerging a great country then it cannot ignore the increasing importance of the Internet in communications, education, business and commerce. Already this significance is being reflected by the increase in the number of cybercafés, ISPs and numerous Internet based businesses in the country. The Internet has moved the economic and social growth of most developed nations and the same effects are being seen in many developing nations. But as reported by internet world stats, Internet penetration in Nigeria stands at only 7.2 percent. Though very poor for a country priding itself as the giant of Africa with a huge population of more than 150 million people, its potential for growth is massive and thus positive.

There are many challenges on its way to growth: some of these challenges are

Some Obstacles affecting internet operations in Nigeria includes:

1.) Lack of power supply – There is limited and unsteady power supply in the country when compared to the rate of supply in other parts of the world. This has served as one of the major problems hindering Internet growth in the country. The government has done virtually nothing to ensure steady power supply. To access the internet fully and steadily, a steady power supply is a criterion, which unfortunately has not been tackled, and this has led to a limited use of the internet by many who desire to use it. With very unreliable power in the country, operators are left with no choice but to acquire and rely constantly on generators; expending huge gallons of fuel daily, in order to distribute internet to offices, homes, or cafes for subscribers. The problem of power situation in the country has further compounded the problem of the few ISPs and Cyber cafes operating in the country.

2.) Poverty – Nigeria is a developing country consisting mainly of poverty-ravaged people who cannot afford three square meals a day. There is high cost of feeding, education, and health in the land. The poor ones will rather not waste their little income to access the internet when they have their priority as food, clothing and shelter. Many cannot afford to own a PC at

home, even the fairly used ones being shipped from developed countries; talk less of engagement in internet connectivity for research and exploration.

3.) High level of ict illiteracy – “Illiteracy is a defect,” says the white man’s adage. The high rate of illiteracy in the country is another ugly trend that is hindering the growth of internet use in Nigeria and this has contributed to the decline in the use of this super facility. People are ignorant of what the internet is all about, its uses, applications and benefits in solving their problems. Most people have not seen a computer before, not to talk of understanding internet concepts. Although it must be said that ICT knowledge in Nigeria is growing and being embraced, yet there remain a large majority of rural population that has not seen direct extensive benefits of the internet.

4.) High cost of getting internet – Internet doesn’t come cheap in Nigeria. One of the reasons for the high cost of connecting to the internet is the costs involved in using satellites due to a lack of cable infrastructure. Most Internet traffic in Nigeria often times needs to be routed by satellite via North America and Europe Website.

5.) Low speed and time-consuming internet – Out of 10 million estimated Internet users in Nigeria, broadband users are few. High-speed broadband requires fiber optic cables. The situation should, however, improve as there are a number of projects underway to connect African countries with the rest of the world through undersea cables which is hoped will eventually lead to affordability and fast internet speed.

6.) Increasing competition – Traditional Internet Service Providers are beginning to face increasing competition from Internet Services offered by mobile operators. The introduction of 3G services in the country has enabled many Nigerians to access the internet through mobile devices. Meanwhile, access to Telecom Services isn’t necessarily reliant on new cabling. While only 4% of people in the country have fixed telephone lines, 30% have cell phones.

7.) Lack of internet ready pcs – Nigerians are not provided with Internet-ready PCs as done in the advanced countries. Though Zinox Computers, a computer manufacturer and a leading CDMA (**code division multiple access** is a channel access method used by various radio communication technologies). Telecom Operator are joining forces to make this available, and other efforts are currently being made, much of this still remain in the

depression.

8.) Yahoo-Yahoo – Another of Nigeria's Internet problems is the growing internet culture among Nigerian youths representing more than 80 per cent of Nigeria's population – Young minds due to lack of awareness, low number of Nigerian website on the internet, lack of Nigerian specific e-payment system and poor e-learning culture through cyber cafes only to check their email accounts, visit pornographic sites, shop online with stolen credit cards and so many other negative vices that has come to be known as **Yahoo-Yahoo!**

Despite all of these daunting challenges, Nigeria is pursuing a path of reforms and modernization. It has the raw materials -- natural resources and educated talent -- to become a world player. And its fledgling Internet industry presents a tremendous opportunity to build bridges with other countries